

Supplementary Table S1

Selection frequency of all 79 genes that entered the top-20 panel at least once across 60 resamples (10 cross-validation folds + 50 repeated 80/20 splits). Genes in the reported 20-gene panel are marked "yes". Frequencies are out of 60.

Gene	Freq /60	In panel
RDH11	60	yes
C8ORF85	60	yes
TRPS1	60	yes
MCTP2	60	yes
ACSM2B	58	yes
TCF21	56	yes
HNF1B	48	yes
KLK3	47	no
HKDC1	46	no
NKX2-1	45	yes
SFTA3	41	yes
KLK2	34	yes
FZD5	32	no
SCGB2A2	31	yes
SCTR	30	no
EN1	30	no
SLC45A3	29	yes
FOXF1	27	yes
IHH	25	yes
EFHD1	24	no
PAX8	23	yes
LMX1B	21	no
OVOL2	20	yes
SFTPA2	20	no
MET	19	yes
MREG	18	no
PRAC	15	no
SCGB3A2	14	no
TFAP2A	13	yes
C15ORF21	13	no
HOXB13	12	yes
C20ORF56	11	no
NAPSA	11	no
GATA3	10	no
GALNT12	9	yes
C9ORF167	9	no
SFTPB	9	no
BEND4	8	no
NKX3-1	7	no
ARHGAP6	6	no

Gene	Freq /60	In panel
SHH	6	no
TARP	5	no
KLK4	5	no
PAX2	5	no
THRSP	4	no
IRX5	4	no
TMEFF2	4	no
TBC1D9	4	no
SFTPA1	4	no
CHRNA2	3	no
HHLA2	3	no
SFTPC	3	no
PRSS22	3	no
SALL1	2	no
PBX1	2	no
SLC4A4	2	no
STEAP2	2	no
EPDR1	2	no
ACPP	2	no
TMEM140	2	no
FOLH1	2	no
CPNE4	2	no
PRLR	2	no
ESM1	1	no
ESRP1	1	no
TMEM30B	1	no
CUX2	1	no
EMX2OS	1	no
SFTPD	1	no
CUBN	1	no
SEL1L3	1	no
CRABP2	1	no
SLC30A4	1	no
HAND2	1	no
FLJ45983	1	no
SLC3A1	1	no
SLC6A13	1	no
ACSM3	1	no
C20ORF151	1	no